

JKBOSE

2026 CLASS 11TH

VECTOR

LECTURE- 05

PHYSICS

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Topics To Be Covered

1 Revision ✓

Vector

10 min

Revision.

One Shot

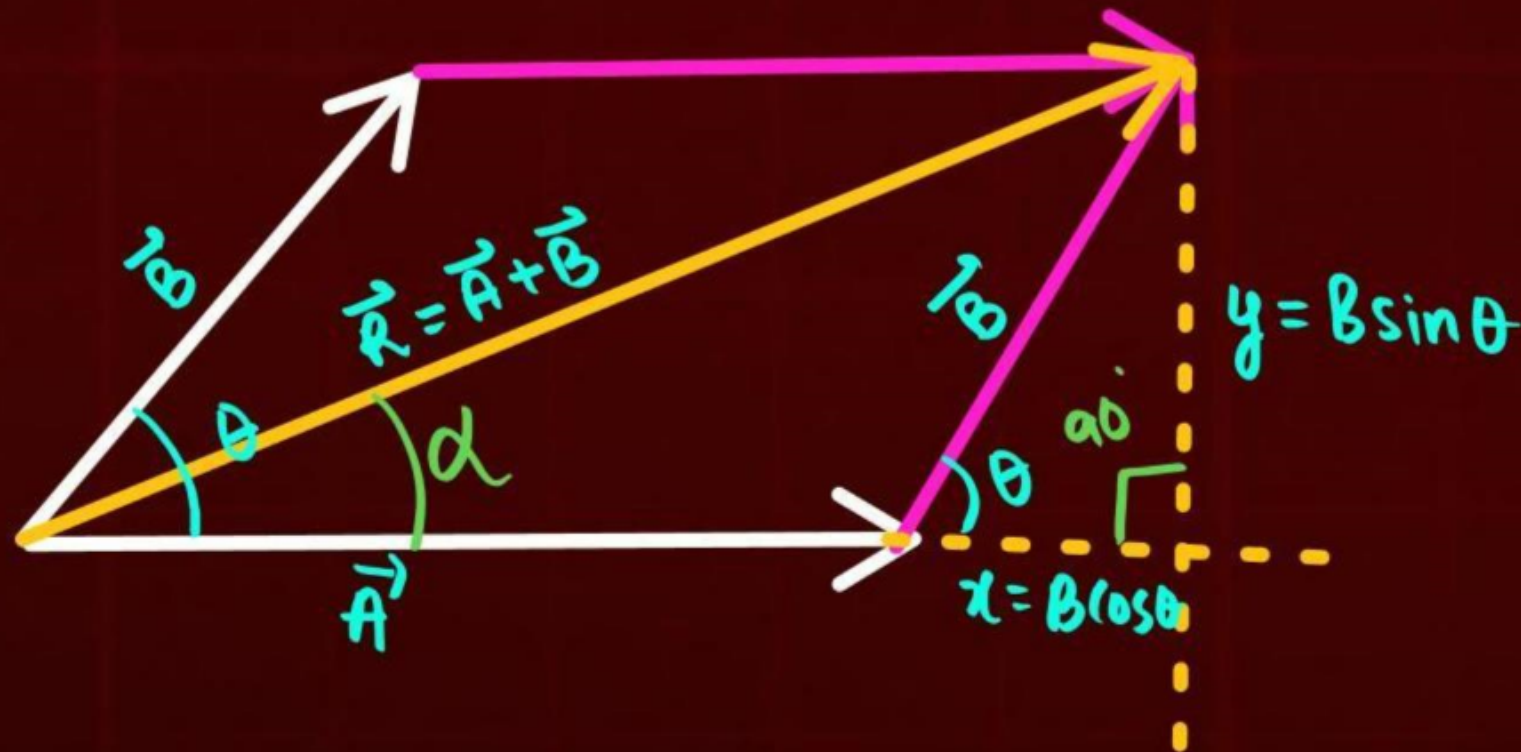
*"4 dec. Dekh k Ahna"

Recap from 2nd Lecture

Parallelogram law of vector Addition

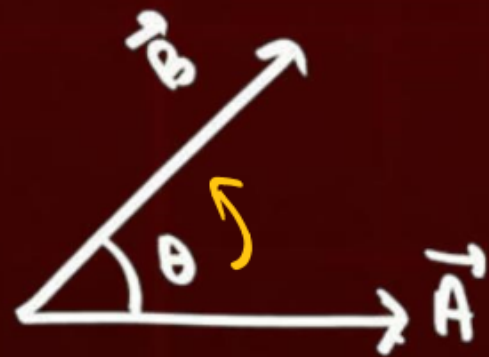
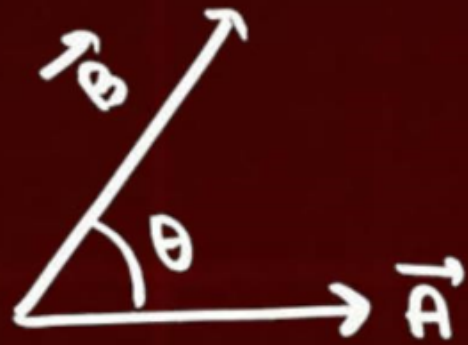
$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

$$\tan\alpha = \frac{B\sin\theta}{A + B\cos\theta}$$



Recap - ③

Cross Product.



(i) Multiplication with integer

$$\vec{A} \times n = n\vec{A}$$

$$\vec{A} \times -n = -n\vec{A}$$

$$\vec{B} \times \vec{A} = \underbrace{BA \sin \theta}_m \underbrace{\hat{n}}_i$$

$$\vec{A} \times \vec{B} = AB \sin \theta \hat{n}$$

Properties of Cross Product.

- ① It is a vector product.
- ② Direction is given by Screw Rule

$$\textcircled{3} \quad AB \sin 90^\circ = AB \times 1 = \underline{AB} \text{ (max)}$$

$$\textcircled{4} \quad AB \sin 0 = AB \times 0 = 0 \text{ (min)}$$

$$\textcircled{5} \quad \vec{A} \times \vec{B} \neq \vec{B} \times \vec{A} \text{ do not show commutative property}$$



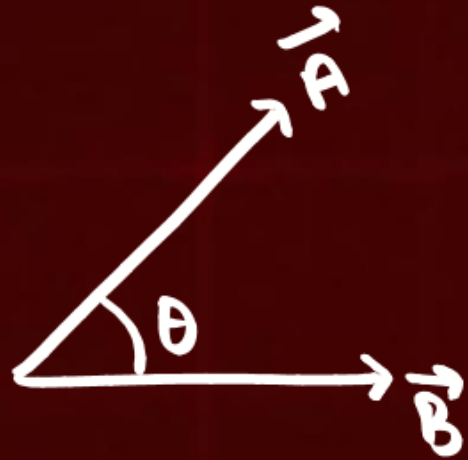
⑥ Distributive property

$$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$$

Recap from 4th Lecture

Dot product of two vector.

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$



$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$\vec{B} \cdot \vec{A} = BA \cos \theta$$

Properties of Dot product.

① $\vec{A} \cdot \vec{B} = AB \cos \theta$ Scalar product.

② no direction ✓

③ $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$ show commutative P

④ $\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$

⑤ $\vec{A} \cdot \vec{B} = AB \cos \theta$
 $= AB \cos 0 = AB \times 1 = AB$ ✓

⑥ $\vec{A} \cdot \vec{B} = AB \cos \theta$ ✓
 $= AB \cos 90 = AB \times 0 = 0$

⑦ $\vec{A} \cdot \vec{B} = AB \cos \theta = AB \cos 180$
 $= AB \times -1 = -AB$

Thank
You